Republic of Rwanda
Ministry of Education

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ETEAI501

ARTIFICIAL INTELLIGENCE INTEGRATION INTO ELECTRONIC DEVICES

Competence: Integrate Artificial Intelligence into electronic devices

RQF Level: 5

Learning Hours
90 hours

Credits: 9

Sector: Technical service

Trade: Electronics and Telecommunication

Module Type: Specific

Curriculum: TSVETE5001-TVET Certificate V in Electronics and Telecommunication

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Issue Date: May, 2024

2024-25

<table>
<tr><td>Purpose statement</td><td colspan="6">This module describes the knowledge, skills and attitude required to integrate artificial intelligence into electronic devices
It is intended to be pursued by learner at TVET certificate level V in Electronics and Telecommunication. At the end of this module, the learner will be able to Perform pre-AI training model activities, Train IA model and Deploy AI.</td></tr>
<tr><td>Learning assumed to be in place</td><td colspan="6">▪ Produce electronic smart devices circuit board
▪ Perform chip programming
▪ Produce electronic smart devices enclosures
▪ Apply python for machine learning</td></tr>
<tr><td rowspan="6">Delivery modality</td><td colspan="2">Training delivery</td><td>100%</td><td colspan="2">Assessment</td><td>Total 100%</td></tr>
<tr><td colspan="2">Theoretical content</td><td>30%</td><td rowspan="4">Formative assessment</td><td>30%</td><td rowspan="4">50%</td></tr>
<tr><td colspan="2">Practical work:</td><td rowspan="3">70%</td><td rowspan="3">70%</td></tr>
<tr><td>• Group work and presentation</td><td>30%</td></tr>
<tr><td>• Individual work</td><td>40%</td></tr>
<tr><td colspan="3"></td><td colspan="2">Summative Assessment</td><td>50%</td></tr>
</table>

Knowledge	Skills	Attitude
✓ Understanding of AI Technologies ✓ Hardware Compatibility ✓ Data Collection and Preprocessing ✓ Model Development and Training ✓ Integration with Device Software ✓ Real-time Inference and Optimization ✓ User Experience Design ✓ Security and Privacy Considerations ✓ Basic of electronic fundamentals ✓ Digital electronics	✓ AI and Machine Learning (ML) Fundamentals ✓ Programming Languages ✓ Embedded Systems and IoT ✓ Hardware Integration ✓ Data Processing and Management ✓ Real-time Inference Optimization ✓ Testing and Validation ✓ Computer skills	✓ Innovative ✓ Honest ✓ Accountability ✓ Self-motivated ✓ Time management ✓ Humble ✓ Creative ✓ Responsible ✓ Flexible ✓ integrity ✓ Goal oriented ✓ Self-confident ✓ Motivated ✓ Energetic ✓ Able to work independently ✓ Strong moral character ✓ Personal hygiene/grooming ✓ Time management ✓ Open-minded ✓ Organized ✓ Maintain health ✓ Positive work ethics ✓ Teamwork and Collaboration ✓

Elements of Competency and Performance Criteria

Elements of competence	Performance criteria
1. Perform pre-AI training model activities	1.1. AI project is appropriately documented in reference to the problem to be solved and AI documentation process
	1.2. AI project related Data are properly collected based on data categories, AI-Data collection techniques and Data pre-processing operations
	1.3. Data file is accurately created according to the Data formats, and file creation process
	1.4. AI dataset is effectively imported based on appropriate development tools, and types of dataset
2. Train IA model	2.1 Training Dataset is appropriately separated from Testing data according to the AI data splitting standard, accuracy parameters
	2.2 AI model is adequately trained according to Machine leaning model training procedures and algorithms
	2.3 AI Model performance is properly evaluated according to AI model test strategies and model accuracy improving techniques
	2.4 Trained model file is correctly exported as per model deployment file formats and AI file libraries
3. Deploy AI	3.1 Appropriate embedded hardware is properly selected according to embedded system board features and specifications
	3.2 Trained Model is accurately Optimized refer to embedded platform specifications and model optimization techniques
	3.3 Optimized model file is properly imported into existing program source code based on used Integrated Development Environment (IDE), AI model integration tools and AI model integration process

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform pre-AI training model activities 2. Train IA model 3. Deploy AI.
4. **Learning outcome 1: Perform pre-AI training model activities**	**Learning hours: 25 hours**

Indicative content

- **Description of AI concepts**
  - ✓ Definitions of key terms
  - ✓ Historical
  - ✓ Importance
- **Documentation of AI project**
  - ✓ **AI project source of information**
  - ✓ **AI vs Problems to be solved**
    - Social
    - Economical
    - technical
  - ✓ AI documentation process
- **Collection of AI project related Data**
  - ✓ AI data categories,
  - ✓ AI-Data collection techniques
  - ✓ Data pre-processing(cleaning, filtering, and Data transformation)
- **Creation of AI Data file**
  - ✓ Data formats
  - ✓ File creation process
- **Import of AI dataset**

- ✓ Development tools
  - Data import software
    - Environment software (anaconda, Visual Studio Code)
  - DATASET Importing techniques
- ✓ Types of dataset

Resources required for the learning outcome	
Equipment	Computer , projectors, camera, mobile phone
Materials	Papers, marker pens, flips chart, pens, chalks,
Tools	Software (anaconda navigator , spyder , VS Code,….)
Facilitation techniques	▪ Brainstorming ▪ Research ▪ presentation ▪ Demonstration ▪ Practical exercise ▪ Trainer guided ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 2: Train IA model	Learning hours: 35 hours
Indicative content	
• **Data splitting** ✓ Training Dataset ✓ Testing data ✓ Dataset Splitting standards Splitting ratio	

- Overfitting
- Underfitting
- Unbiased Evaluation
- Data Variability

✓ Accuracy parameters

- Accuracy
- Precision
- Recall
- F1 score
- Confusion Matrix
- Root mean square error (RMSE)
- Mean absolute Error (MAE)

- **Train AI model/ machine learning model**
  - ✓ Description of Machine learning model
    - Definition of AI model
    - Importance of Model Training
    - Types of machine learning
  - ✓ AI model training procedures
  - ✓ AI model training algorithms
    - Selection of algorithm to train a model
- **Evaluate AI Model performance**
  - ✓ **AI model testing**
    - Types of ML Testing
    - Evaluation Metrics for ML Models
    - Ethical Considerations in ML Testing
  - ✓ **Model accuracy improving techniques**
    - Data Quality Improvement
    - Data Augmentation
    - Feature Engineering
    - Model Selection/ Tuning or Try different algorithm
    - Hyperparameter Tuning

• **Export Trained model as a file** ✓ File formats ✓ AI file libraries

Resources required for the learning outcome	
Equipment	Computer, projectors, camera, mobile phone
Materials	Papers, marker pens, flips chart, pens, chalks
Tools	Software (anaconda navigator, spyder, VS Code….)
Facilitation techniques	▪ Brainstorming ▪ Research ▪ presentation ▪ Demonstration ▪ Practical exercise ▪ Trainer guided ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 3: Deploy AI	Learning hours: 30 hours
Indicative content	

- **Selection of embedded hardware**
  - ✓ Hardware specification and features.
    - Processor
    - Memory
    - Power requirement
    - On-board accelerators (NPU/GPUs)
    - I/0 Interfaces
    - OS/RTOS supported by hardware for ML framework compatibility
    - Supported embedded hardware pricing
  - ✓ Microcontroller and single board computers for ML
    - Arduino Nano 33 BLE Sense
    - ESP-Eye
    - ESP32-CAM
    - Raspberry Pi Boards
    - Nvidia Jetson Nano

- **Optimize Trained Model to embedded platform**
  - ✓ The reason behind model optimization
    - Memory constraints
    - Power limitation
    - Small Computational resources at the edge
  - ✓ Model optimization techniques
    - Quantization
    - Pruning
    - Clustering

- **Deploy Optimized model file is properly**
  - ✓ Imported AI model into existing program source code
    - Compile model to C/C++ code or python
    - Import a compiled model into your source code
  - ✓ AI model integration tools
    - Integrated Development Environment (IDE)
    - Edge Impulse
    - PyTorch
    - TensorFlow
  - ✓ AI model integration process

Resources required for the learning outcome	
Equipment	Computer, projectors, camera, mobile phone, Microcontrollers,
Materials	Papers, marker pens, flips chart, pens, chalks
Tools	Software (anaconda navigator, spyder, VS Code, Pytorch, Edge impulse)
Facilitation techniques	▪ Brainstorming ▪ Research ▪ presentation ▪ Demonstration ▪ Practical exercise ▪ Trainer guided ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment
Integrated/Summative assessment	

Integrated situation

XYZ Company, located in Kigali city, has faced several security breaches in their store rooms. The challenge is to implement a Smart Surveillance System that can intelligently differentiate between thieves (people) and pets (dogs, cats, rats). The system should activate alarms only when a human intruder is detected, thus avoiding false alarms caused by pets.

Areas to be Monitored:

- Store room entrances
- Storage racks

Task

Smart Surveillance System to Differentiate Thieves from Pets

Resources

Tools	Software (anaconda navigator, spyder, VS Code, Pytorch, Edge impulse)
Equipment	• Computer with Arduino IDE and Python environment • ESP32-CAM modules • Breadboards and jumper wires • MicroSD cards • USB cables • Power supply units • Router for Wi-Fi connection
Materials/ Consumables	Papers, marker pens, flips chart, pens, chalks

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
	1.1. AI project is appropriately documented in reference to the problem to be	AI project is well documented			5
		AI documentation processes are respected			5

1. Perform Pre-AI training model activities (40%)	solved and AI documentation process				
	1.2. AI project related Data are properly collected based on data categories, AI-Data collection techniques and Data pre-processing operations	Data collection techniques are respected			5
		Data are well collected			5
		Data preprocessing stage are well performed ❖ Cleaning ❖ Filtering ❖ Transformation			10
	1.3. Data file is accurately created according to the Data formats, and file creation process	Data formats are respected during creation of a file			5
	1.4. AI dataset is effectively imported based on appropriate development tools, and types of datasets	Importation flow of dataset is respected.			5
2. **Train AI model (50%)**	2.1. Training Dataset is appropriately separated from Testing data according to the AI data splitting standard, accuracy parameters	Splitting ratio are followed based to training and test dataset			4
		Overfitting is checked and resolved properly			2
		Underfitting is checked and resolved properly			2
		Bias is evaluated			2
	2.2 AI model is adequately trained according to Machine leaning model training procedures and algorithms	Model training procedures are followed			5
		ML algorithm is selected according to the problem to be solved.			5
		Accuracy is checked			5

	2.3. AI Model performance is properly evaluated according to AI model test strategies and model accuracy improving techniques	Precision is checked			2
		Recall is checked			2
		F1 score is checked			2
		Confusion Matrix is checked			2
		Model improvement is performed			5
	2.4. Trained model file is correctly exported as per model deployment file formats and AI file libraries	Libraries have been well selected to export the model			5
		File format has been selected according to the device to use for deployment.			7
3. Deploy AI (10%)	3.1. Appropriate embedded hardware is properly selected according to embedded system board features and specifications	Hardware is selected according to the problem to be solved			2
	3.2. Trained Model is accurately Optimized refer to embedded platform specifications and model optimization techniques	Model optimization techniques are applied.			2
	3.3. Optimized model file is properly imported into existing program source code based on used Integrated Development Environment (IDE), AI model integration tools and AI model integration process	AI model is compiled to C/C++			2
		AI model integration tools are selected			2
		AI model is correctly integrated to the electronic device.			2
Total marks					**100**
Percentage Weightage					100%
Minimum Passing line % (Aggregate):					70%

List of abbreviations

AI - Artificial Intelligence

RQF - Regulated Qualifications Framework

TVET - Technical and Vocational Education and Training

IDE - Integrated Development Environment

NPU - Neural Processing Unit

GPU - Graphics Processing Unit

OS - Operating System

RTOS - Real-Time Operating System

ML - Machine Learning

RMSE - Root Mean Square Error

MAE - Mean Absolute Error

CAT - Continuous Assessment Test

ESP32 - A series of low-cost, low-power system on a chip microcontroller with integrated Wi-Fi and dual-mode Bluetooth

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Glossary:

Activity: Activities include releases, events, and deployment plans that you develop, start, and complete with the product.

Application environment: A user-defined collection of resources that hosts an application. These application environments refer to environments that are created to be in the product.

Application process: A process that is associated with an application. Unlike a component or generic process, an application process is created from application-level steps.

Application: One or more computer programs or software components that provide a function in direct support of a specific business process or processes. See also application server.

Architecture: The internal structure of a computer system or a chip that determines its operational functionality and performance.

Control processor: The processor in a processor array machine that issues the instructions to be executed by all the processors in the processor array. Alternatively, the control processor may perform tasks in which the processors in the array are not involved, e.g., I/O operations or serial operations.

Deployment: The activities used to deliver a software project to a deployment target. Typically, you run deployments for each stage of your release lifecycle, ending with the production stage when the software becomes generally available.

Duration: The time a task takes to run. Duration is measured from the time a task starts until it is resolved. When you create some task types, you can estimate their expected duration. Duration is reported in minutes.

Environment: A collection of resources that identify the components that can be deployed by the parent application and the agents that do the work.

Events: Release-related activities that are applied to releases and tracked with a calendar. You can use events to organize your releases and other time-dependent activities, such as holidays and blackouts.

Initiative: An action to take for the change occurred.

Insights: A set of related features such as metric data dashboards, reports, and the application portfolio.

Lifecycle: The phases in a release. A lifecycle is a template for the stages of work in a release.

Plugin: A separately installable software module that adds functionality to an existing program, application, or interface.

Resource: A user-defined construct that is based on the architectural model of IBM Urban Code Velocity. A resource represents a deployment target.

Role: A job function that identifies the tasks that a user can perform and the resources to which a user has access. A user can be assigned one or more roles. Processors have direct access to all the memory in the system. Because of technological limitations on shared bandwidth generally, not more than about 16 processors share a common memory.

Task: Represents a business-meaningful activity that has the starting and ending points and a measurable duration. Durations are used to estimate deployment times. You add tasks to deployment plans. When you run a deployment, you complete the tasks in the plan

User: A representation of an account on the server. Users can be members of teams and groups. Users can be created in IBM Urban Code Velocity server or import users from an external authentication realm, including an LDAP, Active Directory, or SSO provider

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PYTHON FOR MACHINE LEARNING

ETEPM501

Apply python for machine learning

Competence

RQF Level: 5

Learning Hours 70

Credits: 7

Sector: TEECHNICAL SERVICES

Trade: ELECTRONICS AND TELECOMUNICATION

Module Type: GENERAL

Curriculum: TSVPP5001 TVET Certificate V in ELECTRONICS AND TELECOMUNICATION

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2024-25

Issue Date: May, 2024

Purpose statement	This specific module describes the skills, knowledge and attitude required to Apply Python Programming. This module is intended to prepare learners pursuing TVET Level 5 in electronics and Telecommunication. Upon completion of this module, the learner will be able to Prepare python environment, write python program and apply python libraries for data analysis.				
Learning assumed to be in place	**N/A**				
Delivery modality	**Training delivery**		**100%**	**Assessment**	**Total 100%**
	Theoretical content		30%	30%	50%
	Practical work:				Group project and presentation Individual project /Work
	• Group project and presentation	30%			
	• Individual project /Work	40%	70%	Formative assessment 70%	
			Summative Assessment		50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Prepare python environment	1.1 Python tools are properly selected in accordance with computer operating system
	1.2 Python tools are properly installed in accordance with Operating System
	1.3 Python installation is successfully tested based on output of python version command
2. Write basic python program	2.1. Python concepts are effectively applied in accordance with recommended programming practices.
	2.2. Control structures are properly applied in line with efficient coding guidelines.
	2.3. Functions are properly applied based on structured programming techniques.
	2.4. Collections are properly applied according to effective data management practices.
	2.5. File handling is properly performed in line with secure and efficient file operations.
	2.6. OOP concepts are properly applied in accordance with object-oriented programming principles.
	2.7. Python date and time concepts are applied according to accurate and efficient time management standards.
	3.1. Numerical libraries are properly used based on precise mathematical operations requirements.

3. Apply python libraries for data analysis.	3.2. Data manipulation libraries are properly applied in accordance with effective data processing and transformation techniques.
	3.3. Data visualization libraries are properly used according to clear and informative data presentation standards.

Knowledge, Skills, and Attitude

Knowledge	**Skills**	**Attitude**
✓ Description of Python libraries ✓ Description of Python language features ✓ Description of Python syntax and semantics ✓ Describing concepts of Python programming	✓ Being Engaged in Problem-Solving ✓ Enhancement of Creativity and innovation skills ✓ Conduct analysis and System ✓ API development	✓ Being Quick learner to achieve the required result. ✓ Being rapid during work execution ✓ Have Trustworthy to details ✓ Being flexible ✓ Indicate team spirit while working with others. ✓ Establish Honesty in your daily activities. ✓ Attention to details while carrying out programming ✓ Take responsibility.

Learning outcomes	At the end of the module the learner will be able to: 1. Prepare python environment 2. Write basic python program 3. Apply object-driven in python
Learning outcome 1: Prepare python environment	**Learning hours: 15**

Indicative content

- **Python programming overview**
  - ✓ Definition
  - ✓ Benefits
  - ✓ Characteristics of python
- **Selection of Python tools**
  - ✓ Applications of python
    - Data science
    - Software development
    - Automation
    - Data analytics
  - ✓ Identifications of python tools
- **Installation of Python tools**
  - ✓ Identification of computer system requirements
    - Hardware requirement
    - Software requirement
  - ✓ Install python software tools
  - ✓ Configure python virtual environment

- **Testing python installation**
  - ✓ Run python version command
  - ✓ Check python interpreter
  - ✓ Test package manager

	Resources required for the learning outcome
Equipment	Computer, Projector
Materials	Internet
Tools	Text editor, Python(latest and stable version), IDE
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises
Formative assessment methods	• Written assessment • Oral questions • Performance assessment

Learning outcome 2: Write basic python program **Learning hours:20**	**Learning hours: 15**

Indicative content

- **Applying python basic concepts**

  ✓ Data Types

  ✓ Variables

  ✓ Comments

  ✓ Operators

- **Applying python control structures**

  ✓ Conditional Statements

  ✓ Looping Statements

  ✓ Use of Jump Statements
    - Break
    - Continue
    - Pass

- **Appling functions in Python**

  ✓ Definition of Function

  ✓ Types of functions
    - built-in function
    - user defined function

  ✓ Create a function
    - Arguments
    - Default parameter value
    - Passing a list as an argument
    - Calling a function

  ✓ Apply special purpose functions
    - Lambda

- Apply Python Generators
- Apply Python Closures
- Apply Python Decorators
- Recursive function
- Higher-order function

- **Applying Python Collections**

✓ **Description of collection types**

- Lists
- Tuples
- Dictionaries
- Sets
- Frozen Set
- ChainMaps
- Deques

✓ Specialized tools from the collection's module

- Counter
- OrderedDict
- defaultdict

✓ Perform common operations

- Adding and Removing Elements
- Accessing and Iterating Over Elements
- Filtering and Sorting
- Set Operations and Counting
- Stack and Queue Operations

- **File handling**

✓ File Handling libraries

- os
- pathlib
- shutil

- ✓ Practice to read file
    - Open file
    - Read file permission
- ✓ Perform write/create file
    - Create a new file
    - Write to existing file
- ✓ Perform delete file
    - Remove file
    - Delete folder
- ✓ Apply python best practices
    - Readability and Style
    - Use of Built-in Features
    - Efficiency and Memory Usage
    - Error Handling and Testing

- **Applying OOP Concepts**

- ✓ Object
- ✓ Python class
- ✓ Inheritance
- ✓ Polymorphism
- ✓ Encapsulation

- **Applying python Date and time concepts**

- ✓ Description of date and time
    - Datetime
    - Dateutil
    - Arrow
    - Pendulum
    - Python-tzdata
- ✓ Set Time zones
- ✓ Formatting and parsing

✓ Perform relative timedeltas

Resources required for the indicative content	
Equipment	Computer, Projector
Materials	Internet
Tools	• Text editor, Python (latest and stable version), IDE
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises
Formative assessment methods	• Written assessment • Oral questions • Performance assessment
Learning outcome 3: Apply python libraries for data analysis.	**Learning hours: 35**

Indicative content

- **Description of Python Libraries**
- **Python standard libraries**

✓ Math

✓ Datetime

✓ Random

- Produce random numbers and sequences
- Randomly shuffle elements in a list
- Draw random samples from a population or list.

- **Python numerical calculation libraries**

✓ NumPy

- Initialize and manipulate arrays
- Perform element-wise operations and matrix manipulations

✓ SciPy

- Apply advanced mathematical functions from submodules
- Use special functions
- Perform scientific computations

- **Data manipulation libraries**

✓ Pandas

- Generate and manipulate dataframe and series objects
- Perform data cleaning and transformation operations
- Conduct data analysis and aggregations

- **Data plotting libraries**

✓ Seaborn

- Create statistical plots such as histograms, box plots, and violin plots
- Adjust aesthetics and themes to enhance visual appeal

- Use built-in functions to visualize relationships and distributions in datasets

✓ Matplotlib

- Generate basic plots
- Modify plot elements such as titles, labels, legends, and colors.
- Export plots to various file formats

Resources required for the indicative content	
Equipment	Computer, Projector
Materials	internet
Tools	Text editor, Python (latest and stable version), IDE, Jupyter notebook
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises
Formative assessment methods	• Written assessment • Oral question • Performance assessment

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PYTHON PROGRAMMING FUNDAMENTALS

GENPP**501**

APPLY PYTHON PROGRAMMING FUNDAMENTALS

Competence

RQF Level: 5

Learning Hours 100

Credits: 10

Sector: ICT and MULTIMEDIA

Trade: NETWORKING AND INTERNET TECHNOLOGY

Module Type: GENERAL

Curriculum: ICTNIT5001 TVET Certificate V in NETWORKING AND INTERNET TECHNOLOGIES

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<tr><td>Learning assumed to be in place</td><td colspan="6">N/A</td></tr>
<tr><td rowspan="7">Delivery modality</td><td colspan="2">Training delivery</td><td>100%</td><td colspan="2">Assessment</td><td>Total 100%</td></tr>
<tr><td colspan="2">Theoretical content</td><td>30%</td><td rowspan="5">Formative assessment</td><td>30%</td><td rowspan="5">50%
Group project and presentation
Individual project /Work</td></tr>
<tr><td colspan="2">Practical work:</td><td rowspan="3">70%</td><td rowspan="3">70%</td></tr>
<tr><td>• Group project and presentation</td><td>30%</td></tr>
<tr><td>• Individual project /Work</td><td>40%</td></tr>
<tr><td colspan="3"></td><td colspan="2">Summative Assessment</td><td>50%</td></tr>
</table>

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Prepare python environment	1.1 Python tools are properly selected in accordance with computer operating system

	1.2 Python tools are properly installed in accordance with Operating System
	1.3 Python installation is successfully tested based on output of python version command
2. Write basic python program	2.1 Python concepts are effectively applied based on python standards
	2.2 Control structures are properly applied in accordance with python standards
	2.3 Functions are properly applied in accordance with python standards
	2.4 Collections are properly applied in accordance with python standards
	2.5 File handling is properly performed in accordance with python standards
3. Apply object-driven in python	3.1 OOP Concepts are properly applied in line with python standards
	3.2 Python date and time concepts are applied according to python standards
	3.3 Libraries are properly used in accordance with python standards
	3.4 System automation is properly applied based on specific task

Knowledge, Skills, and Attitude		
Knowledge	**Skills**	**Attitude**

• Describing Algorithm and data structure • Description of Python libraries • Description of Python language features • Description of Python syntax and semantics • Describing concepts of Python programming	• Being Engaged in Problem-Solving • Enhancement of Creativity and innovation skills • Conduct analysis and System • API development	• Being Quick learner to achieve the required result. • Being rapid during work execution • Have Trustworthy to details • Being flexible • Indicate team spirit while working with others. • Establish Honesty in your daily activities. • Attention to details while carrying out programming • Take responsibility.

Learning outcomes	At the end of the module the learner will be able to: **1.** Prepare python environment **2.** Write basic python program **3.** Apply object-driven in python

Learning outcome 1: Prepare python environment	Learning hours: 15

Indicative content

- **Selection of Python tools**
  - ✓ Python programming overview
    - Definition
    - Benefits
    - Characteristics of python
  - ✓ Applications of python
    - Data science
    - Software development
    - Automation
    - Data analytics
  - ✓ Identifications of python tools
- **Installation of Python tools**
  - ✓ Identification of computer system requirements
    - Hardware requirement
    - Software requirement
  - ✓ Install python software tools
  - ✓ Configure python virtual environment
- **Testing python installation**
  - ✓ Run python version command
  - ✓ Check python interpreter
  - ✓ Test package manager

Resources required for the learning outcome	
Equipment	Computer, Projector
Materials	Internet
Tools	Text editor, Python(latest and stable version), IDE
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises
Formative assessment methods	• Written assessment • Oral questions • Performance assessment
Learning outcome 2: Write basic python program	**Learning hours: 45**
Indicative content	

- **Applying python basic concepts**

  ✓ Data Types

  ✓ Variables

  ✓ Comments

  ✓ Operators

- **Applying python control structures**

  ✓ Conditional Statements

  ✓ Looping Statements

  ✓ Use of Jump Statements

  - Break
  - Continue
  - Pass

- **Appling functions in Python**

  ✓ Definition of Function

  ✓ Types of functions

  - built-in function
  - user defined function

  ✓ Create a function

  - Arguments
  - Default parameter value
  - Passing a list as an argument
  - Calling a function

  ✓ Apply special purpose functions

  - Lambda

- Apply Python Generators
- Apply Python Closures
- Apply Python Decorators
- Recursive function
- Higher-order function

- **Applying Python Collections**

✓ Description of collection types

- Lists
- Tuples
- Dictionaries
- Sets
- Frozen Set
- ChainMaps
- Deques

✓ Specialized tools from the collections module

- Counter
- OrderedDict
- defaultdict

✓ Perform common operations

- Adding and Removing Elements
- Accessing and Iterating Over Elements
- Filtering and Sorting
- Set Operations and Counting
- Stack and Queue Operations

- **File handling**

✓ File Handling libraries

- os
- pathlib
- shutil
- pandas

- ✓ Practice to read file
  - Open file
  - Read file permission
- ✓ Perform write/create file
  - Create a new file
  - Write to existing file
- ✓ Perform delete file
  - Remove file
  - Delete folder
- ✓ Apply python best practices
  - Readability and Style
  - Use of Built-in Features
  - Efficiency and Memory Usage
  - Error Handling and Testing

Resources required for the indicative content	
Equipment	Computer, Projector
Materials	Internet
Tools	• Text editor, Python (latest and stable version),IDE
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises

Formative assessment methods	• Written assessment • Oral questions • Performance assessment
Learning outcome 3: Apply object-driven in python	**Learning hours: 40**

Indicative content

- **Applying OOP Concepts**
  - ✓ Object
  - ✓ Python class
  - ✓ Inheritance
  - ✓ Polymorphism
  - ✓ Encapsulation
- **Applying python Date and time concepts**
  - ✓ Description of date and time
    - Datetime
    - Dateutil
    - Arrow
    - Pendulum
    - Python-tzdata
  - ✓ Set Time zones
  - ✓ Formatting and parsing
  - ✓ Perform relative timedeltas
- **Applying Python Libraries**
  - ✓ Description of Python Library
  - ✓ Python standard library
    - Matplotlib
    - Numpy
    - Pandas
  - ✓ Use of libraries
    - Importing Libraries

- Accessing Functionality
- Understanding Scope according to the name space

- **Automation of application post deployment tasks**

✓ Identification of tasks to automate
- Database migrations
- Configuration file updates
- Service restarts
- Testing and verification
- Logging and notifications

✓ Identification of tasks to be prioritized
- Repetitive
- Time-consuming
- Error-prone
- Critical for deployment speed

✓ Selection of a Python Automation Library
- Fabric
- Ansible
- SaltStack
- Boto3
- vSphere Automation SDK for Python For VMware vSphere environments.
- Consider factors

✓ Develop the Python Script
- Use library functions
- Structure logically
- Logging and output

✓ Integrate script with Deployment Process
- Trigger method to initiate the Python script post-deployment
- Direct execution after deployment completion

- Integration with CI/CD pipelines
- Scheduled execution at specific intervals
- Implement security measures to restrict script access and control sensitive information.

✓ Testing and Monitoring

- Thorough testing
- Monitor script logs execution
- Refine and improve

Resources required for the indicative content	
Equipment	Computer, Projector
Materials	internet

Tools	Text editor , Python(latest and stable version), IDE , Jupyter notebook
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises
Formative assessment methods	• Written assessment • Oral question • Performance assessment

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Ministry of Education

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NITML501

MACHINE LEARNING APPLICATION

APPLY MACHINE LEARNING FUNDAMENTALS

Competence

RQF Level: 5

Learning Hours 120

Credits: 12

Sector: ICT AND MULTIMEDIA

Trade: SOFTWARE DEVELOPMENT

Module Type: Specific

Curriculum: ICTSWDML5001 TVET Certificate V in SOFTWARE DEVELOPMENT

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Issue Date: August 2024

2024-25

<table>
<tr><td>Purpose statement</td><td colspan="5">This specific module describes the skills, knowledge and attitude required to Apply Machine Learning Fundamentals. This module is intended to prepare students pursuing TVET Level 5 in software development. Upon completion of this module, the learner will be able to Apply Data Pre-processing, Develop Machine Learning Model and Perform Model Deployment</td></tr>
<tr><td>Learning assumed to be in place</td><td colspan="5">Python programming, Mathematical analysis statistics and probability</td></tr>
<tr><td rowspan="6">Delivery modality</td><td colspan="2">Training delivery</td><td>100%</td><td colspan="2">Assessment</td><td>Total 100%</td></tr>
<tr><td colspan="2">Theoretical content</td><td>30%</td><td rowspan="4">Formative assessment</td><td>30%</td><td rowspan="4">50%</td></tr>
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<tr><td colspan="2"></td><td colspan="2">Summative Assessment</td><td>50%</td></tr>
</table>

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Apply Data Pre-processing	1.1 Environment is properly prepared based on system requirements.
	1.2 Data is properly manipulated based on the python libraries functionalities.
	1.3 Visualization results are properly interpreted based on its statistical analysis.
	1.4 Data cleaning is appropriately performed based on the provided dataset.
2. Develop Machine Learning Model	2.1 Machine Learning algorithm is properly selected based on the characteristics of the dataset.
	2.2 Machine Learning models are properly trained based on a training set of data.
	2.3 Machine Learning model performance is properly evaluated based on appropriate evaluation metrics
	2.4 Hyperparameters are properly finetuned based on evaluation results

3. Perform Model deployment	3.1.	Deployment methods are clearly selected based on the requirements.
	3.2.	Model file is properly integrated in the system based on the Deployment method (RESTful API guidelines.)
	3.3.	Prediction responses are accurately delivered to the Clients based on the model insights.

Knowledge, Skills, and Attitude

Knowledge	Skills	Attitude
• Description of machine Learning concepts	• Setting up a Machine learning environment • Extracting Dataset • Training Machine Learning model • Integration of Machine Learning model • Problem-Solving skills Creativity and innovation skills	• Quick learning • Creativity • Adaptability • Critical thinking

Learning outcomes	**At the end of the module the learner will be able to:** 1. Apply Data Pre-processing 2. Develop Machine Learning Model 3. Perform Model Deployment
Learning outcome 1: Apply Data Pre-processing	**Learning hours: 30**
Indicative content	
• **Description of Machine learning concepts** ✓ Machine learning overview Definition Machine learning life cycle Machine Learning applications Advantages and disadvantages	

- Difference between machine learning, artificial intelligence, and deep learning
- ✓ Types of Machine Learning
  - Supervised
  - Unsupervised
  - Semi-supervised
  - Reinforcement
- ✓ Machine Learning tools

- **Preparing Machine Learning environment**
  - ✓ Installation of Python
  - ✓ Installation of Tools
  - ✓ Environment Testing
- **Data Collection and Acquisition**
  - ✓ Description of key terms
    - data
    - Information
    - dataset
    - Data warehouse
    - Big data
  - ✓ Identification of Source of data
    - IoT Sensors
    - Camera
    - Computer
    - Smartphone
    - Social data
    - Transactional data
  - ✓ Description of 6 V's of Big Data
    - Volume
    - Variety
    - Velocity
    - Veracity
    - Value
    - Variability
  - ✓ Description of Types of data
    - Structured data
    - Semi-structured data
    - Unstructured data
  - ✓ Gathering Machine Learning dataset

- **Interpret Data Visualization**
  - ✓ Description of data Visualization tools
    - Matplotlib
    - Seaborn
    - Plotly
    - Tableau

- Power BI
- ✓ Use Types of Data Visualization
  - Scatter Plots
  - Line Plots
  - Bar Charts
  - Histograms
  - Box Plots
  - Heat map
- ✓ Applying data Visualization Best Practices
- ✓ Interpreting visualizations results
  - Patterns and Trends
  - Context and Background
  - Correlations and Relationships

- **Perform Data cleaning**
  - ✓ Data cleaning overview
    - Definition
    - Purpose
    - Steps
  - ✓ Description of Characteristics of quality Data
    - Accuracy
    - Completeness
    - Consistency
    - Relevance
    - Validity
  - ✓ Data cleaning for inconsistencies rectification
    - Importance of data Cleaning
    - Data cleaning Techniques
  - ✓ Data normalization
    - Importance of data normalization
    - Data normalization Techniques
  - ✓ Data transformation
    - Importance of data transformation
    - Data transformation Techniques
    - Types of Data transformation

Resources required for the learning outcome

Equipment	Computer, Projector, Storage Devices
Materials	Internet

Tools	Datasets, Python Distribution (CPython, Anaconda), Python IDEs (Jupyter Notebook, PyCharm, Visual Studio Code, or Spyder), Cloud jupyter notebook platform
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises • Trainer guided
Formative assessment methods	• Written assessment • Oral questions • Performance assessment

Learning outcome 2: Develop Machine Learning Model	**Learning hours: 50**

Indicative content

- **Description of Machine learning algorithms and applications**
  - ✓ Supervised Learning Algorithms
    - Linear Regression
    - Logistic Regression
    - Decision Trees
    - Random Forest
    - Support Vector Machine (SVM)
    - k-Nearest Neighbors (KNN)
    - Naive Bayes
  - ✓ Unsupervised Learning Algorithms
    - K-Means Clustering
    - Principal Component Analysis (PCA)
    - Hierarchical Clustering

- Anomaly Detection

✓ Semi-supervised Learning Models

✓ Reinforcement Learning

- Q-Learning
- Deep Q-Networks (DQN)

✓ Neural Network Architectures

- Feedforward Neural Networks
- Neural Networks (Artificial Neural Networks)
- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs)

- **Selection of machine learning algorithm**

✓ Identify problem

- regression
- classification
- clustering

✓ analyse type of data in a given dataset

- Independent variable
- Dependent variable

✓ Resource analysis

- computational power
- memory limitations

✓ choose machine learning algorithm to be used

- **Train machine learning model**

✓ Load a dataset

- Using pandas
- Using Numpy
- Using Scikit-Learn
- Using Seaborn
- Using Requests and io

✓ Split dataset

- train set
- test set

- validation set

✓ Initialize model

✓ Fit the training data into a model

- **Evaluation of machine learning model**

✓ prediction of result

- on test data
- on new data (unseen data)

✓ Visualize predictions

✓ Analyse evaluation metrics

- Accuracy
- precision
- recall
- F1 score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R Squared score
- Adjusted R Squared score

✓ Model interpretation

- **Tuning Hyperparameters**

✓ Define Hyperparameters search space

✓ Choose performance metric

✓ Perform hyperparameter search

✓ Evaluate performance

- Get the best parameters
- Get the best model
- predict on validation data
- apply evaluation metrics on validation data
- apply evaluation metrics on test data

✓ Resolve Potential Bias

- Underfitting
- Overfitting

Resources required for the indicative content

Equipment	Computer, Projector, Storage Devices.
Materials	Internet
Tools	Datasets, Python Distribution (CPython, Anaconda), Python IDEs (Jupyter Notebook, PyCharm, Visual Studio Code, or Spyder), Machine Learning Libraries (NumPy, Pandas, Scikit-Learn, TensorFlow or PyTorch, Matplotlib, Seaborn, Keras, SciPy), Cloud platform.
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises • individualization
Formative assessment methods	• Written assessment • Oral questions • Performance assessment

Learning outcome 3: Perform Model Deployment	**Learning hours: 40**

Indicative content

- **Selection of model deployment method**
  - ✓ Description of model deployment methods
    - Definition
    - Benefits
  - ✓ Identification of system specifications
    - Types of application (web application, mobile app, standalone program, Embedded system)
    - Technology (programming languages and frameworks)
  - ✓ Identification of model specifications
    - Size of the dataset
    - Memory limitations
    - Computing power
    - Format of the model file(Scikit-Learn, TensorFlow SavedModel, ONNX, PyTorch PT)
- **Integration of model file**

- ✓ Integration goal
  - Predictions/Insights
  - Generate content
  - Data analysis
- ✓ Compatibility
- ✓ Interpret API endpoint usage
  - Model serving method
  - Loading strategy
- ✓ Integrate with existing systems
- ✓ Identification data format: Establish the format for input data sent to the API
  - JSON
  - Form data
- ✓ Implement communication
  - Use HTTP requests
  - HTTP Responses to interact with existing system
- ✓ Testing thoroughly deployment
- ✓ Deploy to production
- ✓ Monitor performance
- ✓ Track API requests, response times, and model accuracy.

- **Delivering Prediction to the clients**
  - ✓ Integrating the API into application
  - ✓ Formatting the predictions
  - ✓ Handling errors

Resources required for the indicative content

Equipment	Computer, Projector, Storage Devices.
Materials	Internet
Tools	Datasets, Python Distribution (CPython, Anaconda), Python IDEs (Jupyter Notebook, PyCharm, Visual Studio Code, or Spyder), Machine Learning Libraries (NumPy, Pandas, Scikit-Learn, TensorFlow or PyTorch, Matplotlib, Seaborn, Keras, SciPy), Cloud platform, web application framework, web server
Facilitation techniques	• Group discussion • Trainer guided • brainstorming • Practical exercises
Formative assessment methods	• Written assessment • Oral question • Performance assessment

Integrated assessment

ABZ Company, a pioneering real estate software firm headquartered in Kigali, Rwanda, has been a market leader since its establishment in 2016. Their platform facilitates the seamless buying and selling of houses, catering to a substantial user base. However, recently, the company has been inundated with complaints from users regarding discrepancies in listed house prices. Users have expressed concerns that some sellers are setting prices that do not accurately reflect the value of the properties or the local market conditions.

Upon investigating these claims, ABZ Company discovered that certain sellers were listing properties either to gauge their current market value or due to inadvertent errors. Recognizing the importance of addressing this issue to maintain user trust and ensure fair transactions, ABZ Company has made the strategic decision to leverage machine learning technology.

In response to these challenges, ABZ Company is seeking to hire a skilled machine learning engineer to develop a robust pricing model and deploy it using APIs. This model will utilize various factors to accurately predict the price range of houses listed on the platform. By implementing such a solution, ABZ Company aims to enhance transparency, fairness, and accuracy in property pricing, thereby improving the overall user experience.

This work should be done in 8 hours

NB: Dataset and web application have to be provided by a trainer.

Resources

Equipment	Computer, Storage Devices
Materials	Python Distribution (CPython / Anaconda), Python IDEs (Jupyter Notebook / PyCharm / Visual Studio Code, or Spyder), Machine Learning Libraries (NumPy, Pandas, Scikit-Learn, TensorFlow or PyTorch, Matplotlib, Seaborn, Keras, SciPy), web application framework, web server
Tools	Internet, Dataset, web application

Assessable outcomes	(Assessment criteria based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Apply Data Pre-processing	1.1 The environment is properly configured according to specific functionalities.	Functionalities are specified			5
		The environment is configured.			5
(30%)	The data is properly acquired based on the problem requirements	Data are acquired from the given data source			10

	Data is appropriately pre-processed to obtain a dataset according to the intended model.	Data are pre-processed			20
Learning Outcome 2: Develop Machine Learning Model (50%)	The features are properly engineered based on data analysis requirements.	Data features are engineered			10
		Model features are engineered			10
	Machine learning algorithms are properly selected based on the dataset and problem requirements.	Machine learning models to be used are identified			5
	Model parameters are properly selected according to the algorithm.	Model parameters are selected.			5
	The model is correctly trained based on the algorithm and parameters.	Models are trained and tested.			20
Learning Outcome 3: Perform Model deployment **(20%)**	Metrics are appropriately selected to measure performance accuracy according to the trained model	Metrics are selected.			2
		Metrics are implemented			8
		Model parameters are tuned.			5
	The machine learning model is properly deployed into production using relevant deployment best practices.	The selected fit model is saved.			2
		The model is deployed to the web app.			3
Total marks					**100**
Percentage Weightage					**100%**
Minimum Passing line % (Aggregate): 50%					

References:

1. Brownlee, J. (2016). Master Machine Learning Algorithms - Discover how they work.

2. Eric, H., Sinayobye, O., Masabo, E., Ufitinema, C., Murangira, T., Rwibasira, P., . . . Gaurav, B. (2023). An Intelligent System-Based Coffee Plant Leaf Disease Recognition Using Deep Learning Techniques on Rwandan Arabica Dataset. Technologies, 23.

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NITIW401

IoT WEB APPLICATION DEVELOPMENT USING PHP

Develop IoT Web Application Using PHP

Competence

RQF Level: 4

Learning Hours 100

Credits: 10

Sector: ICT and Multimedia

Trade: Network and Internet Technologies

Module Type: Specific

Curriculum: ICTNIT4001 TVET Certificate IV in Networking and Internet Technologies

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1200 **Issue Date:** September, 2023

<table>
<tr><td>Purpose statement</td><td colspan="6">This specific module describes the skills, knowledge and attitude required to develop an IoT web application. This module is intended to prepare student pursuing TVET Level 4 in Networking and Internet Technologies.
Upon completion of this module, the learner will be able to develop an API using PHP, develop User Interface, Integrate API Endpoints with User Interface, develop a web application and document an IoT web application using PHP.</td></tr>
<tr><td>Learning assumed to be in place</td><td colspan="6">▪ Database Development</td></tr>
<tr><td rowspan="6">Delivery modality</td><td colspan="2">Training delivery</td><td>100%</td><td colspan="2">Assessment</td><td>Total 100%</td></tr>
<tr><td colspan="2">Theoretical content</td><td>20%</td><td rowspan="4">Formative assessment</td><td>20%</td><td rowspan="4">50%</td></tr>
<tr><td colspan="2">Practical work:</td><td rowspan="3">80%</td><td rowspan="3">80%</td></tr>
<tr><td>Group project and presentation</td><td>30%</td></tr>
<tr><td>Individual project /Work</td><td>50%</td></tr>
<tr><td colspan="2"></td><td colspan="2">Summative Assessment</td><td>50%</td></tr>
</table>

Elements of Competence and Performance Criteria

Elements of competence	Performance criteria
1. Develop API using PHP	1.1 Web application requirements are properly analysed based on organization requirements
	1.2 PHP environment is appropriately prepared based on web application requirements
	1.3 PHP concepts are properly applied based on PHP standards
	1.4 API CRUD endpoints are properly developed based on IoT system requirements.
	1.5 API endpoints are properly secured based on PHP data security standards.
	1.6 API endpoints are effectively tested based on IoT system requirements
	1.7 Application Programming Interface (API) are properly documented according to its functionality
2. Develop user interface	2.1 User Interface (UI) requirements are properly identified based on IOT system requirements
	2.2 HTML tags are appropriately used based on wireframe
	2.3 CSS is properly applied to the HTML tags based on system layout
3. Integrate API Endpoints with User Interface	3.1 Application Programming Interface (API) endpoints are correctly interpreted according to API documentation.
	3.2 Application Programming Interface (API) data is properly presented on the user interface in line with http communication protocol
	3.3 IOT web application is effectively documented based on system functionality

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe PHP concepts ✓ Identify system requirements ✓ Describe API concepts ✓ Knowledge of CRUD ✓ Identify UI requirements ✓ Security Practices	✓ Install software ✓ Develop and secure Application Programming Interface Endpoints ✓ Develop UI ✓ Display API data on User Interface ✓ Interpret API Endpoints ✓ Document IoT Web Application	✓ Exploratory ✓ Critical thinking ✓ Innovativeness ✓ Persistence ✓ Self confidence ✓ Team work spirit ✓ Adaptability

Course content

Learning outcomes	**At the end of the module the learner will be able to:** 1. Develop Application Programming Interface (API) using PHP. 2. Develop user interface 3. Integrate Application Programming Interface (API) Endpoints with User Interface

Learning outcome 1: Develop Application programming interface (API) using PHP.	Learning hours: 40

Indicative content

- **Analysis of IoT Web application requirements**
  - ✓ Identify system requirement.
    - Collect the data.
    - Select the required devices while developing and displaying
    - Identify source of data (from sensors, from database, from APIs)
    - Describe types of system architecture
  - ✓ Functional requirements
  - ✓ Non-functional requirements
  - ✓ Code management
    - Version control and collaboration
    - Automation and continuous delivery
    - Code minimization and optimisation

- **Prepare PHP environment**
  - ✓ Description of key terms
    - PHP (purpose, characteristics)
    - Interpreter
    - Compiler
    - Open Source
    - Web Server
    - Database (DBMS, MySQL, NoSQL)

- Browser
- Text Editor
- Integrated Development Environment(IDE)

✓ Installation of environment tools

- Install Apache, mysql, php (xampp, mamp, wamp,lamp)
- Install Browser
- Install Text Editor
- Install Integrated Development Environment (IDE)

✓ Configure Apache, MySQL, php (xampp, mamp, wamp,lamp)

✓ Test the environment using phpinfo() function

- **Identification of PHP concepts**

✓ Tags

✓ Variable

- Naming rules
- Types
- Scopes

✓ Super Global variables

✓ Operators

✓ Data types

✓ Variable scope

✓ Constants

✓ Comment

✓ String concatenation

✓ Control structures

- Conditional statements
- Iterating statement

- Switch statement
- Jump statements

✓ Arrays

- Indexed array
- Associative array
- Multidimensional array

✓ Functions

- Built-in functions
- User-defined functions
- Anonymous functions
- Calling function

✓ File handling

- Opening a file
- Reading a file
- Writing a file
- Closing a file
- Deleting a file

✓ Data Validations

- **Develop Application Programming Interface CRUD Endpoints**

✓ Description of Application Programming Interface concepts

- Definition
- Types (Representational state Transfer (REST), Simple Object Access Protocol(SOAP)
- API working principles

- Request methods
- Endpoint

✓ Application of data formats (JSON)

✓ Execute CREATE query

- Register IoT device
- Validate data from IoT device
- Store data in database

✓ Execute READ Query

- Retrieve device information
- Fetch data from external Application Programming Interface
- Check device status
- Access alert records

✓ Execute UPDATE Query

- Update device information
- Modify sensor settings
- Update device firmware
- Update access permission
- Update alert and notification settings

✓ Execute DELETE Query

- Remove sensor data
- Delete command history
- Purge inactive devices
- Delete alert configurations

✓ Error Handling

- **Secure API endpoints**
  - ✓ Implementing Application Programming Interface authentication
  - ✓ Apply Application Programming Interface security Best practices
  - ✓ Managing Application Programming Interface security
- **Test API endpoints**
  - ✓ Description of Application Programming Interface testing concepts
    - Testing tools (Postman, Sandbox, PHPUnit, Guzzle, Pest, Dredd-PHP)
    - API testing best practices
    - Benefits of Application Programming Interface testing
    - Testing steps
  - ✓ Types of Application Programming Interface endpoints testing
    - Unit testing
    - Integration testing
    - End to end testing
    - Load testing
  - ✓ Testing of Endpoint accessibility
- **Documentation of the developed API**
  - ✓ Choose documentation tool (Postman/swagger/ApiDoc/Spotlight)
  - ✓ Document API specifications(information)
  - ✓ Document Application Programming Interface Endpoints
    - Include authentication information
    - Include Request parameters
    - Include Response structure
    - Include Error codes and error response formats

➢ Add usage guidelines ✓ Document Application Programming Interface Versioning

Resources required for the learning outcome	
Equipment	▪ Computer ▪ Sensors ▪ Arduino boards ▪ Node MCU boards ▪ Raspberry pi
Materials	▪ Electricity ▪ Internet
Tools	▪ XAMPP ▪ WAMP ▪ LAMP ▪ MAMP ▪ DBMS ▪ MSQL ▪ NoSQL Postman ▪ Swagger ▪ ApiDoc ▪ Spotlight ▪ Sandbox ▪ PHPUnit ▪ Guzzle ▪ Behat ▪ Pest ▪ Dredd-PHP ▪ Google chrome ▪ visual studio

	▪ Sublime text ▪ Notepad++
Facilitation techniques or Learning activity	▪ Demonstration ▪ Individual and group work ▪ Practical exercise ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Learning outcome 2: Develop user interface	**Learning hours: 25**
Indicative content	

- **Identification of UI requirements**
  - ✓ Review System requirements
  - ✓ List key user services
  - ✓ Define user personas
  - ✓ Description of user experience (UX) best practices
    - Define user interface
    - Clear and Intuitive Navigation
    - Consistent Design
    - Use Familiar UI Patterns
    - Choose an attractive colour scheme
  - ✓ Features of a good user interface
  - ✓ Draw Wireframe
- **Use HTML tags**

- ✓ Description of HTML
  - Definition
  - Versions
  - Tag attributes
- ✓ HTML tag categories
  - HTML Structural tags
  - Formatting tags
  - Table tags
  - Form tags
  - Heading tags
  - List tags
  - Media tags
  - Code tags
  - HTML frame tags
  - HTML Comment
  - Grouping tags (div, span)
  - Hyperlink tag
  - Semantic tags

- **Application of CSS**
  - ✓ Description
    - Definition
    - Versions
    - Style types
  - ✓ Use CSS Visual rules

- Font
- Colors and background colors
- Opacity
- Background image

✓ Use CSS Display and positioning

- Relative
- Absolute
- Fixed
- Sticky
- Block

✓ Use CSS Box and Grid model

- Height and width
- Borders
- Border radius
- Padding
- Margin
- Visibility
- Auto

Resources required for the indicative content	
Equipment	▪ Computer ▪ Projector
Materials	▪ Electricity ▪ Internet
Tools	▪ Text editor (Notepad, Notepad++, Sublime text) ▪ Web browser

Facilitation techniques or Learning Activity	▪ Group discussion ▪ Demonstration ▪ Practical exercise ▪ Individual work
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Learning outcome 3: Integrate API Endpoints with User Interface	**Learning hours: 35**
Indicative content	

- **Interpret API Endpoints**
  - ✓ Read description of each endpoint HTTP method
  - ✓ Check URL structure
  - ✓ Check Request parameters
  - ✓ Check the structure of API responses
  - ✓ Check if the API requires authentication
  - ✓ Check error codes and error response format.

- **Use API data on User Interface**
  - ✓ URL requests
    - Endpoints
    - Method
    - Headers
    - Data (or body)

- Status (code and message)

✓ HTTP requests

- POST requests to create records
- GET request to read or get a resource from the server
- PUT and PATCH requests to update records
- DELETE request to delete a resource from a server

✓ Testing and Debugging

✓ Data rendering

- Show Loading States and Progress Indicators
- Preserve Filtering and Sorting
- Preserve Pagination and Infinite Scrolling
- Ensure responsiveness design

- **Document IoT Web Application**

✓ User guide documentation

✓ Report on work done

✓ Provide Bills of quantities

Resources required for the indicative content	
Equipment	▪ Computer ▪ Projector
Materials	▪ Internet ▪ Electricity
Tools	▪ Text editor

	▪ Webserver ▪ Web browser ▪ DBMS
Facilitation techniques or Learning activity	▪ Demonstration ▪ Group discussion ▪ Practical exercise ▪ Individual work
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Integrated/Summative assessment

Integrated situation

ZOMBI Farmers ltd is a cooperative found in Rwanda, Eastern region, Nyagatare District. These farmers grow Maize. Due to prolonged dry seasons that lead to poor yields, the cooperative members adopted an irrigation system that operates manually in shifts. Recently the members realized that some members dodge their shifts which lead to drying of crops due to irregular irrigation. The cooperative decided to hire an IoT Web application developer to develop for them a web application for their irrigation system that will help them to monitor and control the soil moisture and temperature. As an IoT web Application developer, you are required to perform the following task.

1. Create API endpoints to:

    a. Consume weather parameters from irrigation system equipment.

    b. Store soil information to the database

c. Represent data on the user interface.

d. If the soil moisture levels go below 50% the water pump should start watering automatically.

e. Alert the farm manager about soil moisture level through a telephone message

f. Create a configuration interface that will allow user to set up irrigation parameters (soil moisture< = 500 for normal condition and soil moisture>500 for starting watering)

Information:

Provided is a database design.

Database name: ZOMBI

Tables:

FarmData: fd_Id (pk), Moisture_level, temp _degree, date_time

Farmer: f_id(pk), first_name, last_name, sex, phone_number, fd_Id(fk)

This work should be done in 12 hours

Resources	
Equipment	▪ Computer
Tools	▪ Web server ▪ Text editor / IDE ▪ Web browser ▪ Testing tool(telephone) ▪ API Documentation tool

Assessable outcomes	**Assessment criteria (Based on**	**Indicator**	**Observation**		**Marks allocation**
			Yes	**No**	

	performance criteria)				
Learning outcome 1: Develop API using PHP **(30%)**	**1.1:** API CRUD endpoints are properly developed based on IoT system requirements.	**1.** CREATE query is executed			**4**
		2. READ Query is executed			**4**
		3. UPDATE Query is executed			**4**
		4. DELETE Query is executed			**4**
	1.2: API endpoints are properly secured based on PHP data security standards.	**1.** Authentication is implemented			**4**
		2. Security is implemented			**3**
		3. Security is managed			**3**
	1.3: API endpoints are effectively tested based on IOT system requirements	**1.** Testing tool is selected			**3**
		2. Point to point test is performed			**4**
		3. Endpoint access test is performed			**4**
	1.4: API are properly documented according to its functionality	**1.** Documentation tool is selected			**2**
		2. API functionality is specified			**2**

		3. Endpoints are included			**2**
		4. API Version is indicated			**2**
Learning outcome 2: Develop user interface (40%)	**2.1:** UI requirements are properly identified based on IOT system requirements	**1.** Attractive colours are selected			**3**
		2. key user services are identified			**3**
		3. Wireframe is drawn			**4**
	2.2: HTML tags are appropriately used based on wireframe	**1.** Formatting tags are used			**4**
		2. Code tags are used			**4**
		3. Semantic tags are used			**4**
	2.3: CSS is properly applied to the HTML tags based on system layout	**1.** Colours and images are used			**3**
		2. CSS display and positioning techniques are applied			**4**
		3. CSS box or grid model is used			**4**
Learning outcome 3:	**3.1:** API data is properly presented	**1.** create record requests are performed			**3**

Integrate API Endpoints with User Interface (30%)	on the user interface in line with http communication protocol	**2.** Resource is accessed from server			**4**
		3. PUT and PATCH operations are performed			**4**
		4. Delete operations are performed			**3**
	3.2: IoT web application is effectively documented based on system functionality	**1.** user guide is written			**3**
		2. Report is provided			**3**
		3. Bill of quantities is provided.			**3**
Total marks					**100**
Percentage Weightage					**100%**
Minimum Passing line % (Aggregate): 70%					

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D9

Republic of Rwanda
Ministry of Education

RTB | RWANDA TVET BOARD

INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

CCMCS402

Apply Computer Skills

Competence

RQF Level: 4

Learning Hours 30

Credits: 3

Sector: All

Trade: All

Module Type: Complementary Module

Curriculum: All

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Issue Date: May 2023

<table>
<tr><td>Purpose statement</td><td colspan="6">This module describes knowledge skills and attitudes required to apply computer skills. At the end of this module, learners will be able to describe the operating system; Customize the computer features and protect the computer.</td></tr>
<tr><td rowspan="6">Delivery modality</td><td colspan="2">Training delivery</td><td>100%</td><td colspan="2">Assessment</td><td>Total 100%</td></tr>
<tr><td colspan="2">Theoretical content</td><td>30%</td><td rowspan="4">Formative assessment</td><td>30%</td><td rowspan="4">100%</td></tr>
<tr><td colspan="2">Practical work:</td><td rowspan="3">70%</td><td rowspan="3">70%</td></tr>
<tr><td>• Group Work and presentation</td><td>20%</td></tr>
<tr><td>• Individual Work</td><td>50%</td></tr>
<tr><td colspan="2"></td><td colspan="2">Summative Assessment</td><td>N/A</td></tr>
</table>

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe the operating system	**1.1** Operating system is properly described according to its activities
	1.2 Main Operating system components are properly described according to their services
	1.3 Operating system types are properly described according to the particular task
	1.4. Operating system file system is properly described according to their types
2.Customize the computer features	**2.1.** Customer requirements are properly identified for an operating system according to their needs
	2.2. Minimum Hardware Requirements are properly identified for an operating system to be used
	2.3. Operating system is properly customized according to the perspective steps
3. Protect	**3.1.** Software utilities are properly installed according to the manufacturer

computer system	specifications
	3.2. Computer is regularly scanned according to the elimination of virus as per safety standards
	3.3. Computer data are accurate backed up and restored based on OS installation

Course content

Learning outcomes	At the end of the module the learner will be able to:
	1. Describe the operating system 2. Customize the computer features 3. Protect computer system

Learning outcome 1: Describe the operating system	Learning hours: 15
Indicative content	

- **Description of operating system**
  - ✓ Definition of operating system
  - ✓ Features of operating system
  - ✓ Functions of operating system
    - Memory management
    - Processor management
    - Device management
    - File management
    - Security management
    - Coordination other software and users
    - Error detecting aids
    - Job accounting
- **Description of the main components of operating system**
  - ✓ User interface
    - Graphical user interface (**GUI**)
    - Command Line Interface (**CLI**)
  - ✓ Kernel
    - Monolithic kernel

- Microkernel

✓ File management system

- Select group of Adjacent files or directories
- Select group of Non adjacent files or directories
- Duplicate files /folder
- Move /delete files and folders
- Downloading files/folders

- **Description of the types of operating system**

✓ Batch operating system

✓ Distributed operating system

✓ Real Time Operating System

- Hard real time operating system
- Soft real time operating system

✓ Network operating system

✓ Time –sharing operating system

✓ Mobile operating system

✓ Embedded operating system

- **Description of the operating system file system**

✓ File system role

✓ Major types of file systems in Microsoft windows

- File Allocation Table (FAT12, FAT16, FAT32)
- New Technology File System (NTFS)

Resources required for the learning outcome	
Equipment	▪ Computer ▪ Projector ▪ Whiteboard
Materials	▪ Marker ▪ Flipchart

Tools	▪ CD ▪ DVD ▪ Flash driver ▪ Reference books
Facilitation techniques	▪ Individual and group work ▪ Individualized ▪ Trainer guided ▪ Group discussion
Formative assessment methods	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 2: Customize the computer features	**Learning hours: 10**
Indicative content	

- **Identification of the customer requirements for an operating system**
  - ✓ Requirements to be considered
    - Budget constraints
    - Compatibility with current hardware
    - Compatibility with new hardware
    - How the computers will be used?
    - Compatibility with existing applications
    - Types of new applications to be used
- **Identification of the minimum hardware requirements for an operating system**
  - ✓ Basic requirements for installing windows 10 on PC
    - RAM capacity:1Gigabyte (1GB) for 32-bits or 2 Gigabytes (2GB) for 64-bits
    - Hard drive size:16 GB for 32-bits or 32 GB for 64-bits OS
    - Graphics card: DirectX or later with WDDM 1.0 driver
    - Processor: 1GHZ or faster processor or System on a Chip

- Display: 800 x 600

✓ Common hardware upgrades

- **Customization of the operating system**
  - ✓ Windows settings
    - Creating a user account
    - Manage user account
    - Window Personalization
  - ✓ Disk management
    - Disk partitioning
    - Disk formatting
    - Disk clean up
  - ✓ Windows administrative tools
    - Task scheduler
    - Event viewer
  - ✓ Control panel settings

Resources required for the indicative content	
Equipment	▪ Computer ▪ Projector ▪ Whiteboard
Materials	▪ Marker ▪ Flipchart
Tools	▪ CD ▪ DVD ▪ Flash Driver ▪ Reference books
Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Practical exercise

	▪ Individualized ▪ Trainer guided ▪ Group discussion
• **Formative assessment methods**	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 3: Protect computer system	**Learning hours: 5**
Indicative content	

- **Installation of software utilities**
  - ✓ Description of software tools \utilities
    - Anti-virus
    - Anti-malware
    - Anti-spyware
    - Backup software
    - Backup media
  - ✓ Installation of software tools\utilities
    - Anti-virus
    - Anti-malware
    - Anti-spyware
- **Scanning and Eliminating virus**
  - ✓ Description of scan types
  - ✓ Description of scan mode
  - ✓ Virus elimination
    - Elimination by neutralization
    - Elimination by deleting
    - Elimination by quarantine

- **Backing up and restoring of computer data**
  - ✓ Description of backup types
    - Full backup
    - Differential backup
    - Incremental backup
    - Copy backup
    - Mirror backup
    - Daily backup
  - ✓ Selection of common backup devices
    - Tape drive
    - Optical Discs
    - USB flash drivers
    - SD card
    - HDD/SSD
  - ✓ Selection of common restore devices
    - HDD/SSD
    - USB flash Drivers
    - Tape drive
    - SD Card and Optical drives

Resources Required For The Indicative Content	
Equipment	▪ Computer ▪ Projector ▪ Whiteboard
Materials	▪ Markers ▪ Flipchart
Tools	▪ USB/ Flash Driver ▪ HDD ▪ SD Card

	▪ Tape drives ▪ Reference books
Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Practical exercise ▪ Individualized ▪ Trainer guided ▪ Group discussion
Formative assessment methods	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

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Republic of Rwanda
Ministry of Education

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ETECP302

C PROGRAMMING FUNDAMENTALS

APPLY C PROGRAMMING FUNDAMENTALS

Competence

RQF Level: 3

Learning Hours 90

Credits: 9

Sector: Technical Services

Trade: Electronics and Telecommunication

Module Type: General

Curriculum: TSVETE3001-TVET Level 3 in Electronics and Telecommunication

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Issue Date: April 2022

Purpose statement	Electronics and Telecommunication domain is shifting from manual manipulated to more automated installations. To automate equipment, they need to make decisions depending on the inputs that they are reading. The control of these decisions is done through programming of their control parts. At the end of this module, trainees will be equipped with knowledge and skill to describe computer-programming languages, write a C program and run the programming.					
Learning assumed to be in place	N/A					
Delivery modality	**Training delivery**		**100%**	**Assessment**		**Total 100%**
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Apply Computer programming Languages	1.1. Programming languages are properly identified considering their types and application
	1.2. Algorithm is conveniently developed considering the problem to be solved
	1.3. Flowchart is accurately developed based on the logic of question
2. Write C Programming codes	2.1. C programming building blocks are properly described based on programming standards
	2.2. C program structure is property described according to programming standards
	2.3. Condition statements are effectively applied according to the logic of the problem to be solved.
	2.4. Loops are effectively applied according to the logic of the problem to be solved.
	2.5. Functions are effectively applied according to the logic of the problem to be solved.
	2.6. Arrays are effectively applied according to the logic of the problem to be solved.

	2.7. C problems are correctly solved by using C programming techniques
3. Perform Program testing Debugging	3.1. Different program errors are carefully identified according to the programming standards
	3.2. The program is systematically compiled in accordance with the program instructions.
	3.3. The program is tested and errors are completely corrected based on the program and programming instructions.

Course content

Learning outcomes	**At the end of the module the learner will be able to:** 1. Identify computer programming Languages 2. Write C Programming codes 3. Perform Program Debugging
Learning outcome 1: Describe Computer programming Languages	**Learning hours: 15**
Indicative content	

- **Identification of programming Language**
  - ✓ Types of programming languages
  - ✓ Role of programming languages in electronics and telecommunication
  - ✓ Types of C programming IDE
  - ✓ Setting up C programming environment
- **Development of a flowchart**
  - ✓ Flowchart symbols
  - ✓ Steps of developing a flowchart
- **Development of an algorithm**
  - ✓ Types of algorithm
  - ✓ Steps of developing an algorithm

Resources required for the learning outcome

Equipment	Computers, Projector, Projection screen, Printers
Materials	Chalks, Pens, Books, Papers, Flip-chart
Tools	Internet access, C programming IDE
Facilitation techniques and learning activities	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

Learning outcome 2: Write C Programming codes	**Learning hours: 65**
Indicative content	

- **Description of C programming building blocks**
  - ✓ Tokens in C
  - ✓ Data types
  - ✓ Variables
  - ✓ Constants
  - ✓ Operators
- **Description of C program structure**
  - ✓ Preprocessor Commands
  - ✓ Functions
  - ✓ Variables
  - ✓ Statements & Expressions

- ✓ Comments

- **Application of condition statements**
  - ✓ If
  - ✓ Else if
  - ✓ Switch statement
  - ✓ Nested conditions
- **Application of loops**
  - ✓ For
  - ✓ While
  - ✓ Do while
  - ✓ Nested loops
- **Application of functions**
  - ✓ Types of function
  - ✓ Defining a Function
  - ✓ Function Declarations
  - ✓ Calling a Function
  - ✓ Function Arguments
- **Application of Arrays**
  - ✓ Types of arrays
  - ✓ Declaring Arrays
  - ✓ Initializing Arrays
  - ✓ Initializing Arrays
- **Solving a c programming problem**
  - ✓ Draw the problem flowchart
  - ✓ Design the algorithm to solve the problem
  - ✓ Write the C program to solve the problem
  - ✓ Program documentation

Resources required for the indicative content

Equipment	Computers, Projector, Projection screen, Printers
Materials	Chalks, Pens, Books, Papers, Flip-chart
Tools	Internet access, C programming IDE
Facilitation techniques	• Demonstration and simulation • Individual and group work • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

Learning outcome 3: Perform Program Debugging	**Learning hours: 10**
Indicative content	
• **Identification of errors** ✓ Types of errors ✓ Techniques of correcting errors • **Compilation of the C program** ✓ Types of compilers ✓ Compilation techniques • **Test of the C program** ✓ Design of effective test cases ✓ Debugging techniques	
Resources required for the indicative content	

Equipment	Computers, Projector, Projection screen, Printers
Materials	Chalks, Pens, Books, Papers, Flip-chart
Tools	Internet access, C programming IDE
Facilitation techniques	● Demonstration and simulation ● Individual and group work ● Individualized ● Trainer guided ● Group discussion
Formative assessment methods	● Written assessment ● Oral presentation ● Performance assessment ● Product based assessment ● Project based assessment

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Republic of Rwanda
Ministry of Education

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NITIS301

IoT SYSTEM INSTALLATION

INSTALL IoT SYSTEM

Competence

RQF Level: 3

Learning Hours: 100

Credits: 10

Sector: ICT & MULTIMEDIA

Trade: NETWORKING AND INTERNET TECHNOLOGIES

Module Type: Specific

Curriculum: ICTNIT3001 - TVET Certificate III in Networking and Internet Technologies

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Issue Date: August 2022

<table>
<tr><td>Purpose statement</td><td colspan="6">This specific module describes the skills, knowledge and attitude required to install IoT System. This module is intended to prepare student pursuing TVET Level 3 in Networking and Internet Technologies. Upon completion of this module, the learner will be able to prepare IoT system installation, deploy IoT equipment, operate, test and maintain IoT system.</td></tr>
<tr><td rowspan="6">Delivery modality</td><td colspan="2">Training delivery</td><td>100%</td><td colspan="2">Assessment</td><td>Total 100%</td></tr>
<tr><td colspan="2">Theoretical content</td><td>30%</td><td rowspan="4">Formative assessment</td><td>30%</td><td rowspan="4">50%</td></tr>
<tr><td colspan="2">Practical work:</td><td rowspan="3">70%</td><td rowspan="3">70%</td></tr>
<tr><td>Group project and presentation</td><td>20%</td></tr>
<tr><td>Individual project /Work</td><td>50%</td></tr>
<tr><td colspan="2"></td><td colspan="2">Summative Assessment</td><td>50%</td></tr>
</table>

Elements of Competency and Performance Criteria

Element of competence	Performance criteria
1. Prepare IoT system installation	1.1. Installation manual is properly interpreted based on the installation design
	1.2. IoT components are properly identified based on the installation design
	1.3. Tools, materials, and equipment are correctly selected according to the installation requirements
	1.4. Tool, devices, and equipment are accurately calibrated as per manufacturer's standards
2. Deploy IoT equipment	2.1. The workplace is properly set according to the installation design
	2.2. IoT equipment are correctly assembled as per the manufacturer's guides
	2.3. Equipment settings are appropriately customized based on the installation requirements
	2.4. IoT system protection measures are properly applied as per the installation manual

	2.5. Functional testing is correctly performed according to the installation requirements
	2.6. IoT system installation is properly reported based on the installation requirements
3. Operate IoT system	3.1. Types of IoT system are properly identified according to the installation manual
	3.2. IoT system features are properly identified according to the IoT system requirements
	3.3. IoT system is properly monitored according to the IoT system requirements
	3.4. Data backup is correctly performed as per the backup guidelines
	3.5. IoT system status is properly interpreted as per the installation manual
	3.6. IoT system incidents are properly reported as per the monitoring results
4. Test IoT system	4.1.Types of testing are properly identified according to the installation requirements
	4.2.IoT Testing tools are correctly selected based on the type of testing
	4.3.Connectivity testing is correctly performed according to the installation requirements
	4.4.Functional testing is correctly performed according to IoT system installed
	4.5.Performance testing is correctly performed according the IoT system installed
	4.6.Testing report is correctly elaborated based on the test results
5. Maintain IoT system	5.1.Features updates are properly installed based on the manufacture's guidelines
	5.2.IoT system protection measures are properly checked in accordance with the installation manual
	5.3.Preventive Maintenance report is properly elaborated based on the work done

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Prepare IoT system installation 2. Deploy IoT equipment 3. Operate IoT system 4. Test IoT system 5. Maintain IoT system
Learning outcome 1: **Prepare IoT system installation**	Learning hours: **20**

Indicative content

- Introduction to IOT
  - ✓ Description of IoT Fundamentals
    - Definition
    - Benefits
    - Applications
    - Devices/Components
    - Platforms
    - Evolution
  - ✓ Identification of IoT components
    - Things
    - Internet
    - Connectivity
  - ✓ Identification of IoT Tools, Materials and equipment
    - Software
    - Hardware
  - ✓ Identification of IoT Technology Challenges
    - IoT Security
    - Data Privacy
    - IoT Devices Safety
    - Interoperability
    - Environmental Impact
- Identification of IoT Network Technologies
  - ✓ Low Range/Low Power
    - Zigbee
    - Wi-Fi6
    - Bluetooth/LE
    - NFC
    - RFID
    - Ethernet
  - ✓ Large Range
    - LPWAN
    - Cellular

- LoRa
- SigFox
- WiFi HaLow
- 5G
- LTE
- Satellite

- Description of IoT Architecture and protocols
  - ✓ Physical model of IoT
  - ✓ Logical model of IoT
  - ✓ IoT Protocols
- Calibration of tools, equipment and materials
  - ✓ Description of calibration
    - Definition
    - Purpose
  - ✓ Calibration techniques
  - ✓ Calibration procedures
- Interpretation of Manual/Datasheet

Resources required for the learning outcome	
Equipment	Wireless Router, Gadgets, Appliances, Arduino Board
Materials	Arduino Starter Kit, IoT Device Datasheet
Tools	Calibration software, Simulators, Screw drivers, Pliers, Multi-meter
Facilitation techniques	Demonstration and simulation Individual and group work Trainer guided Group discussion
Formative assessment methods /(CAT)	Written assessment Oral presentation

Learning outcome 2: **Deploy IoT equipment**	Learning hours: **20**
Indicative content	

- Performance of workplace Setup
  - ✓ Study the workplace environment
    - Distance
    - Physical obstacles
    - Existing systems (Source of power)
  - ✓ Selection of requirements
    - Tools
    - Equipment
    - Materials

- ✓ Mapping the workplace
- ✓ Setup network connectivity

- Identification of IoT deployment levels
  - ✓ Level-1
  - ✓ Level-2
  - ✓ Level-3
  - ✓ Level-4
  - ✓ Level-5
  - ✓ Level-6
- Assembling of IoT equipment
  - ✓ Manufacturer's guide interpretation
  - ✓ Fixing of IoT equipment
- Configuration of IoT devices
  - ✓ Selection of IoT platform
    - On-premise
    - Cloud based
  - ✓ Installation of IoT applications
    - IoT Device provisioning
    - IoT Management (Future Ready device management eg: remote configuration, Unified Device Management)
  - ✓ Functional testing
    - Speed
    - Latency
    - Jitter
    - QoS
- Implementation of IoT system Security
  - ✓ Setup the security measures
    - Physical
    - Logical
  - ✓ Perform IoT device firmware Update
    - Firmware versioning
    - Firmware control
- Documentation of IoT installation report
  - ✓ Data and facts recording
  - ✓ Report generation

Resources required for the indicative content	
Equipment	Modem, Access Point, Computer, microcontrollers, Arduino Starter Kit
Materials	Internet connectivity, pipes, cable trunks, electrical tapes, screws, Labelling tags
Tools	Software platforms, simulation software, tape measure, multi-meter, screwdrivers, drilling machine, hammer, ladder, pliers,

Facilitation techniques	Demonstration and simulation Individual and group work Practical exercise Individualized Trainer guided
Formative assessment methods /(CAT)	Written assessment Oral presentation Performance assessment Product based assessment
Learning outcome 3: **Operate IoT system**	Learning hours: **20**

Indicative content

- Identification of IoT systems Types
  - ✓ Consumer IoT
  - ✓ Commercial IoT
  - ✓ Industrial IoT
  - ✓ Infrastructure IoT (IIoT)
  - ✓ Internet of Military Things (IoMT)
- Implementation of IoT system Backup and restoration
  - ✓ Identification of backup types
    - Full backup
    - Differential backup
    - Incremental backup
  - ✓ Creation of backup
    - Automatic
    - Manual
  - ✓ Restoration of IoT system
- Identification of IoT features
  - ✓ Connectivity
  - ✓ Sensing
  - ✓ Active Engagements
  - ✓ Scale
  - ✓ Dynamic Nature
  - ✓ Intelligence
  - ✓ Energy
  - ✓ Safety
  - ✓ Integration
- Management of IoT device service
  - ✓ Event triggering
  - ✓ Conditional actions
  - ✓ Conditional configurations
- Monitoring of IoT system
  - ✓ Monitoring software
  - ✓ Monitoring parameters

- ✓ IoT system status interpretation
- ✓ Monitoring results interpretation
- ✓ Incident management
  - Definition
  - Lifecycle

Resources required for the indicative content	
Equipment	Modem, Access Point, Computer, microcontrollers, Arduino Starter Kit
Materials	Internet connectivity, pipes, cable trunks, electrical tapes, screws, Labelling tags
Tools	Software platforms, simulation software, tape measure, multi-meter, screwdrivers, drilling machine, hammer, ladder, pliers,
Facilitation techniques	Demonstration and simulation Individual and group work Practical exercise Individualized Trainer guided
Formative assessment methods /(CAT)	Written assessment Oral presentation Performance assessment Product based assessment

Learning outcome 4: **Test IoT system**	Learning hours: **20**
Indicative content	

- Identification of IoT Testing Types
  - ✓ Usability
  - ✓ Reliability
  - ✓ Data integrity
  - ✓ Security
  - ✓ Compatibility
  - ✓ Performance
- Performance of IoT Tests
  - ✓ Functional test
  - ✓ Security test
    - Identification of security testing tools
    - Perform Security test
  - ✓ Connectivity test
    - Identification of connectivity tools
    - Perform Connectivity test
  - ✓ Performance test
    - Identification of performance test

- Perform Performance test
- Elaboration of testing report
  - ✓ Generation of report
  - ✓ Interpretation of report

Resources required for the indicative content	
Equipment	Computer, Multi-meter, Digital Storage Oscilloscope
Materials	Internet connectivity
Tools	Software platforms, JTAG Dongle, Software-defined radio (SDR)
Facilitation techniques	Demonstration and simulation Individual and group work Practical exercise Individualized Trainer guided
Formative assessment methods /(CAT)	Written assessment Oral presentation Performance assessment Product based assessment

Learning outcome 5: **Maintain IoT system**	Learning hours: **20**

Indicative content

- Identification of IoT Maintenance types
  - ✓ Predictive
  - ✓ Preventive
- Application of IoT Software upgrade
  - ✓ Automatic
  - ✓ Manual
- Performance of IoT system security maintenance
  - ✓ Perform IoT action repairs
  - ✓ Perform IoT system protection checks
- Documentation of IoT maintenance report
  - ✓ Data analysing and facts recording
  - ✓ Report generation

Resources required for the indicative content	
Equipment	Computer, Multi-meter, Digital Storage Oscilloscope
Materials	Internet connectivity,
Tools	Software platforms, JTAG Dongle, Software-defined radio (SDR)
Facilitation techniques	Demonstration and simulation Individual and group work Practical exercise Individualized

	Trainer guided
Formative assessment methods /(CAT)	Written assessment Oral presentation Performance assessment Product based assessment

Integrated/Summative assessment

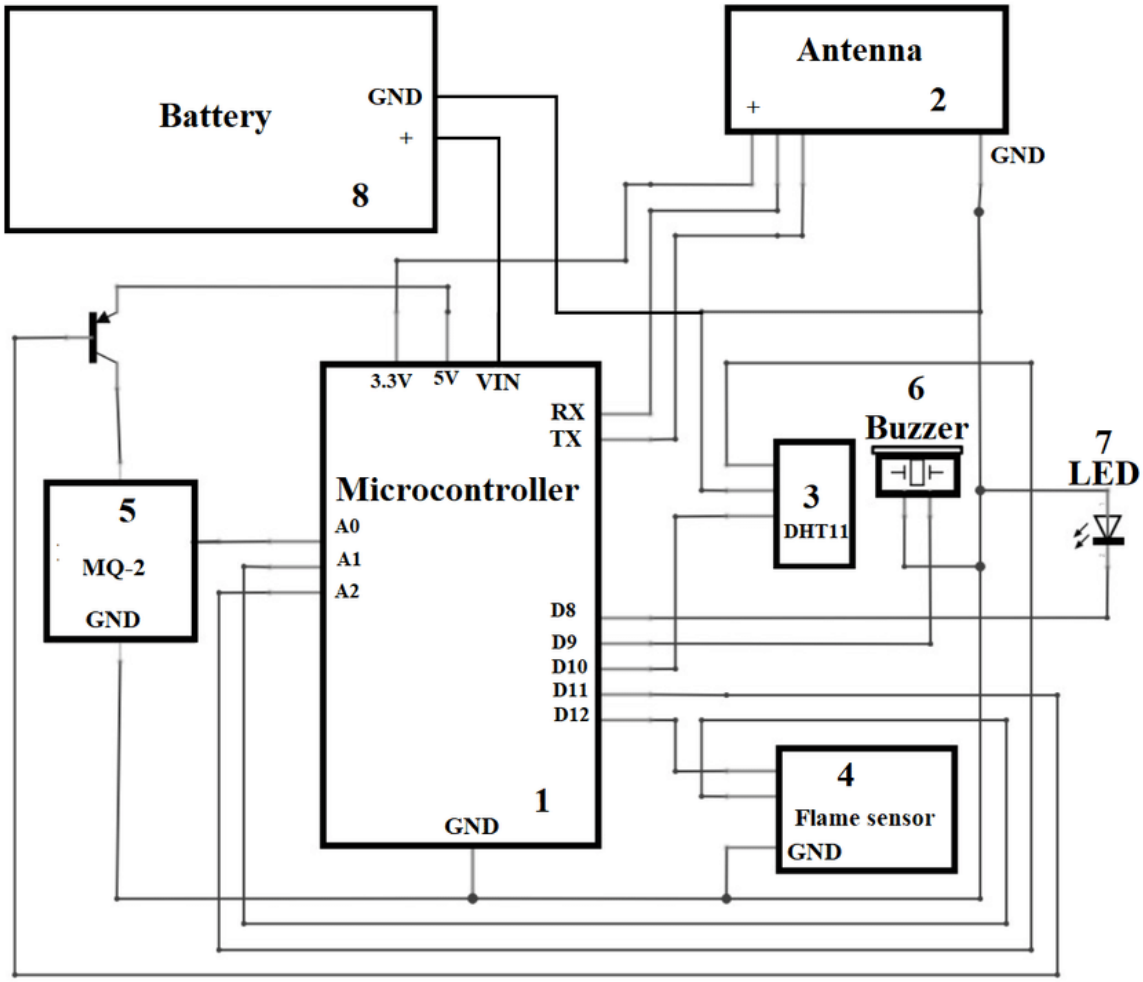
Integrated situation

HH Media Ltd is a networking company located in Kigali City. They provide internet services to public and private institutions.

Institution that works with them have declared a problem concerning how to prevent their premises to be burnt without notice.

After being presented this problem, HH Media Ltd hired an IoT Engineer to develop for them a Fire Alarm IoT Solution that can handle that problem and after developing this solution he presented this diagram as installation design to be followed whenever it is going to be deployed to the institutions.

HH Media Ltd now needs an IoT Installer who will be in charge of assembling and deploying the system, and make sure that a maintenance plan is to be provided at the end. You are given the following diagram to help you install the developed IoT Solution and after the work you have to provide a detailed installation report as well as testing report.



Current Consumption as follows:

Module	Current per hour	Current per second
DHT11 Sensor	0.3mA	0.00008333mA
MQ-2 Sensor	160mA	0.04444mA
Flame Sensor	0.4mA	0.0001111mA
Arduino's microprocessor (sleep-mode)	0.0001mA	0.000000027778mA
Arduino's microprocessor (active mode)	0.3mA	0.00008333mA
XBee ZigBee Antenna (send mode)	33mA	0.0091666mA
XBee ZigBee Antenna (receive mode)	28mA	0.00777778mA
XBee ZigBee Antenna (sleep-mode)	0.0001mA	0,0000002778mA

Device Specifications:
DHT11 Sensor: Temperature & Humidity Sensor (20-95%RH ; 0-50℃; 5V)
MQ-2 Sensor: Smoke/Gas Sensor with Concentration of 200 – 10000ppm
Flame Sensor: IR Based, KY-026
Arduino Board: Arduino Uno
XBee Antenna: ZigBee Antenna Module

This work has to be done within 7 hours. All tools, equipment and materials are provided.

Resources

Tools	Internet Connectivity, Digital Multi-meter
Equipment	Programmed Microcontroller (Arduino Uno)
Materials/ Consumables	Arduino Starter Tool Kit, Flame Sensor, XBee

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Prepare IoT system installation: (20%)	Installation manual is properly interpreted based on the installation design	Installation manual is interpreted			3
		Installation design is interpreted			3
	IoT components are properly identified based on the installation design	IoT components are identified			2
		IoT components are selected			2
	Tools, materials, and equipment are correctly selected according to the installation requirements	Tools, materials, and equipment are selected			3
		IoT equipment are assembled			3
	Tool, devices, and equipment are accurately calibrated as per manufacturer's standards	Tool, devices, and equipment are calibrated			4
Deploy IoT equipment: (25%)	The workplace is properly set according to the installation design	Distance and physical obstacles are identified			3
		Existing system is analysed			3
		The workplace is set			3
	IoT equipment are correctly assembled as per the manufacturer's guides	Manufacturer's guide is interpreted			3
		IoT equipment are fixed			2
	IoT system protection measures are properly applied as per the installation manual	IoT system protection measures are applied			3
	Functional testing is correctly performed according to the	Functional testing is performed			3

	installation requirements				
	IoT system installation is properly reported based on the installation requirements	IoT system installation is reported			4
Operate IoT system: (25%)	IoT system features are properly identified according to the IoT system requirements	IoT system features are properly identified			3
	IoT system is properly monitored according to the IoT system requirements	Monitoring software and parameters are identified			3
		Monitoring software parameters are selected			3
		IoT system is monitored			2
	Data backup is correctly performed as per the backup guidelines	Backup type is identified			2
		Backup is performed			4
	IoT system status is properly interpreted as per the installation manual	Event triggering is identified			2
		Conditional actions are interpreted			3
		IoT device services management are performed			3
Test IoT system: (15%)	Types of testing are properly identified according to the installation requirements	Types of testing are identified			1
	IoT Testing tools are correctly selected based on the type of testing	Testing tools are selected			2
	Connectivity testing is correctly performed according to the installation requirements	Connectivity tools are selected			1
		Connectivity testing is performed			2
	Functional testing is correctly performed according to IoT system installed	Functional testing is performed			2
		Functional testing report is generated			2

	Performance testing is correctly performed according the IoT system installed	Performance testing tool is selected			1
		Performance testing is performed			2
	Testing report is correctly elaborated based on the test results	Testing report is generated			2
Maintain IoT system (15%)	Features updates are properly installed based on the manufacture's guidelines	Preventive Maintenance is performed			7
	IoT system protection measures are properly checked in accordance with the installation manual	Protection measures are performed			5
	Preventive Maintenance report is properly elaborated based on the work done	Preventive Maintenance report is elaborated			3
Total marks					**100**
Percentage Weightage					**100%**
Minimum Passing line % (Aggregate): **70%**					

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Glossary

Actuator: A mechanism that performs a physical task based on input from a connected system.

Application Agents: Help address the lack of overhead for end-to-end, peer-to-peer networking in IoT architecture by their presence in the propagator nodes in an enterprise. They move intelligence to the edge of the network to help manage traffic, allow a real-time response to changing IoT conditions, and provide local client services.

Bluetooth Low Energy (BLE): A wireless personal area network (PAN) aimed at devices with reduced power consumption and cost while maintaining a similar communication range to regular Bluetooth.

Connected Devices: Components that make up the Internet of Things. Many have built-in sensors and/or actuators and collect data to help users or other devices make informed decisions and monitor or affect outside events.

Connectivity: A part of the Edge Layer that serves to ensure that device connectivity doesn't fail if there is a network failure or an unreliable connection.

Edge Gateway: The connecting factor between device analytics and cloud data processing and analytics

Embedded Device/Systems: A computer with a dedicated function within a larger mechanical or electrical system; it is embedded as part of a complete device.

Endpoint Device: An Internet-capable device on a TCP/IP network.

Home Automation: A combination of hardware and software solutions that allow for the control and management of electronics, appliances, and devices within a home.

Industrial Internet: The integration of machine learning, big data technology, sensor data, and machine-to-machine communication automation. This is done with the knowledge that the Internet of Things will be scaled and driven by enterprises. The idea is that smart machines can more accurately capture and communicate data to help corporations find problems sooner and increase overall efficiency.

Integrator: The “tree trunk” of network architecture that performs the big data functions to provide a higher-level analysis of human interaction for near-edge analytics and broader-scope analysis and control.

Internet of Things (IoT): A network of objects (such as sensors and actuators) that can capture data autonomously and self-configure intelligently based on physical world events, allowing these systems to become active participants in various public, commercial, scientific, and personal processes.

Internet Protocol Suite (TCP/IP): The language a computer uses to access the Internet. It consists of a suite of protocols designed to establish a network of networks to provide a host with access to the Internet.

IoT Cloud Platform: A cloud platform that provides a set of services that simplify the integration process between the services provided by cloud platforms and IoT devices. Some platforms include development tools and data analytics capabilities.

IoT Development Board: A board that can be used to prototype and create IoT hardware. There are several boards available on the market with different features.

Lightweight Protocol: Any protocol that has a lesser and leaner payload when being used and transmitted over a network connection.

Long Range Communication Protocols: Used to refer to universal long-range radio frequencies for multi-generation wireless standards such as 2G, 3G, 4G, and4G LTE.

Low-Power Devices: Electronics that have been designed to use less electric power than traditional devices. These are necessary to the future success of IoT because, as sensors become more advanced, devices need to be able to operate for longer periods of time without relying on manual maintenance or loss of data.

Machine-to-Machine (M2M): This refers to a network setup that allows connected devices to communicate freely, usually between a large number of devices; M2M often refers to the use of distributed systems in industrial and manufacturing applications.

Mesh Network: A type of network topology in which a device transmits its own data and also serves as a relay for other nodes by providing the most efficient data path through routers.

Microcontroller (MCU): A small computer on a single integrated circuit designed for embedded applications and used in automatically controlled embedded systems.

Messaging Protocols: The way information is transferred and communicated amongst devices, the cloud, and data storage. Different protocols are used for different results.

Message Queuing Telemetry Transport (MQTT): A lightweight messaging protocol that runs on the TCP/IP protocol. It is designed for communicating with small devices in remote locations with low network bandwidth.

Multi-Agent System: A network of multiple agents which act in an environment and interact or communicate with each other to achieve their design objective.

Near-Field Communication (NFC): A feature based on technical standards that allows devices to establish radio communication with other nearby systems or mobile devices.

Operability: Operability is the measure of how well a software system works when operating in production, whether that is the public cloud, a co-located datacenter, an embedded system, or a remote sensor forming part of an IoT network.

Personal Area Network: A network created through the interconnection of information technology devices within the context of a single user.

Propagator: The “leaves” of the network architecture tree that are serviced by intermediate branch network elements. They manage message routing protocol translation services.

Radio Frequency Identification (RFID): A technology that incorporates electromagnetic coupling and radio frequency to identify objects and persons. It consists of three components: an antenna, transceiver, and transponder.

Real-Time Operating System (RTOS): Designed to guarantee the completion of a task within a certain time constraint. Often used in safety-critical systems and when building IoT devices.

Releasability: The ability to quickly deploy changes to a software system, but also to quickly recover from disaster and adapt to changing technical and business challenges.

Sensor: A device or component that perceives and responds to physical input from the environment.

Sensor Network: A group of sensors with a communications infrastructure intended to monitor and collect data from multiple locations.

Single-Board Computer: A complete computer built on a single circuit board with all the components required of a functional computer.

Store and Forward: A messaging mechanism in which a broker is involved between sender and receiver so that the broker gets ownership of the message from the sender, stores it for reliability, and then delivers the message itself to the receiver.

System on a Chip: An integrated chip that is comprised of electronic circuits of multiple computer components to create a complete device.

Transmission Control Protocol/Internet Protocol (TCP/IP): A basic client/server model communication protocol for the Internet and private networks.

Wearables: Connected devices that can be equipped with different types of sensors and are worn on a person’s body. They are meant to monitor, collect, and quantify data about a person's life and environment, and allow them to interface with that data.

Wi-Fi: A wireless local area network (WLAN) that uses radio waves to provide wireless high-speed Internet and network connections.

ZigBee: An open standard for wireless communication designed to use low-power digital radio signals for personal area networks (PAN); it is used to create networks that require a low data transfer rate, energy efficiency, and secure networking.

Z-Wave: A wireless protocol for home automation that communicates using a low-power radio frequency technology specifically designed for remote control applications.

Abbreviations

5G: 5th Generation

BLE: Bluetooth Low Energy

DHT: Digital Temperature and Humidity

GND: Ground

IIoT: Infrastructure IoT

IoMT: Internet of Military Things

IoT: Internet of Things

JTAG: Joint Test Action Group

LoRa: Long Range

LPWAN: Low Power WAN

LTE: Long Term Evolution

M2M: Machine to Machine

MCU: Microcontroller

MQTT: Message Queuing Telemetry Transport

NFC: Near-Field Communication

PAN: Personal Area Network

QoS: Quality of Service

RFID: Radio Frequency Identification

RTOS: Real-Time Operating System

RX: Receive

SDR: Software-defined radio

TCP/IP: Transmission Control Protocol/Internet Protocol

TX: Transmit

Wi-FI: Wireless Fidelity

Author’s Note Page

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INDUSTRIAL ATTACHMENT PROGRAM (IAP)

ICTIA302 Integrate the workplace

Competence

RQF Level: 3

Learning Hours 200

Credits: 20

Sector: all

Trade: all

Module Type: Specific

Curriculum: **all**

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Issue Date: September 2023

1200

Purpose statement	This module describes the skills, knowledge and attitudes required to integrate the workplace. At the end of this module, participants know how to apply for and present themselves for employment. They demonstrate good time management and show up for work on time. They demonstrate behavior and attitudes that are appropriate for the workplace and understand that workplaces have policies and procedures that need to be followed. They take initiative and responsibility for their own work and know how to work under and respect supervision.					
Delivery modality	**Training delivery**		**100%**	**Assessment**		**Total 100%**
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
				Summative Assessment		N.A

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Apply for internship	1.1. Resources in the community or nationwide are appropriately used to find internship information.
	1.2. Neat CV and an application letter are accurately written.
	1.3. Interview skills are properly demonstrated
2. Demonstrate workplace behavior and attitudes	2.1. Workplace habits and attitudes are appropriately applied.
	2.2. Strategies to manage time effectively are adequately implemented.
	2.3. Personal life at workplace is efficiently managed.
3. Analyze own professional gaps	3.1. Professional gaps in own carrier is properly identified based on labour market demand.
	3.2. Strategies to fill the professional gaps are properly established in accordance with labor market opportunities.
	3.3. Relevant placement is well selected based on identified gaps.
4. Enhance acquired competencies at the workplace	4.1. Tasks assigned are adequately performed according to the agreement with the enterprise.
	4.2. IAP documents are properly prepared based on the work done.
	4.3. Work experience gained during IAP is properly described.
	4.4. Skills transfer based on acquired competences are properly described.

Course content	
Learning outcomes	**At the end of the module the learner will be able to:** 1. Apply for internship 2. Demonstrate workplace behavior and attitudes 3. Analyze own professional gaps in line with industry demands 4. Enhance acquired competencies at the workplace
Learning outcome 1: Apply for internship	**Learning hours: 10**
Indicative content	

- **Resources in the community or nationwide used to find internship information**.
  - ✓ Resources to find internship in the community and at national level
    - Job announcements
    - Newspapers
    - Advertisements
    - Recruitment agencies
    - Internet websites
  - ✓ Identify contact persons
- **Writing on curriculum vitae (C.V) and an application letter**
  - ✓ Differences between a CV and an application letter
  - ✓ Types of application letters
  - ✓ Elements of a well written CV
  - ✓ Elements of an application letter
- **Demonstration of interviewing skills**
  - ✓ Main parts of an interview
  - ✓ Interviewing tips:
    - what to do before
    - what to do during interview
    - what to do after the interview
  - ✓ Typical questions asked during an interview

Resources required for the learning outcome	
Equipment	• Computer, Projector, chalk board, flipchart stand
Materials	• Paper, flipchart

Tools	• Sample of IAP application letter, Sample of Curriculum vitae, Sample of Complain letter, Sample of thanks letter, Reference book Pictures • Power Point presentation on mapping the industry organizational structure, • Sample of organizational structure,
Facilitation techniques	• Individual and group work • Group discussion and presentation • Learning through interaction of guest speakers (Practitioners) • Learning through visits on the relevant institutions • Case study
Formative assessment methods	• Written assessment • Oral presentation

Learning outcome 2. Demonstrate workplace behavior and attitudes	**Learning hours: 10**
Indicative content	

- **Identification of appropriate workplace behaviors and attitudes.**
  - ✓ Appropriate workplace behaviors and attitudes
    - Dress code
    - Time management
    - Respect,
    - Honesty
    - Integrity
    - Work as a team
  - ✓ Work habits
    - Cooperation,
    - Initiative,
    - Courtesy,
    - Constructive
    - Criticism,
    - Supervision
    - Accuracy,
    - Piece of work,
    - Time usage
    - Adaptability
- **Time management.**
  - ✓ Importance of task planning and managing time
  - ✓ Strategies to better manage time
    - Start your day with a clear focus.

- Have a dynamic task list.
- Focus on high-value activities.
- Minimize
- Interruptions.
- Limit multi-tasking.
- Review your day.

- **Managing personal life at workplace.**
  - ✓ Common situations that make it challenging to balance work & personal life
  - ✓ Tips to manage personal and work lives

Resources required for the indicative content	
Equipment	• Projector, computer, Flipchart Stand
Materials	• Flipchart, Markers, Chalks, Paper, stickers, scotch tape.
Tools	• Power Point presentation on creativity and innovation, Reference books
Facilitation techniques	• Individual and group work • Group discussion and presentations • Learning through interaction of guest speakers (Practitioners) • Learning through visits on the relevant institutions • Case study
Formative assessment methods	• Written assessment • Performance assessment • Oral presentation

Learning outcome 3: Analyze own professional gaps in line with industry demands	Learning hours: 10

Indicative content

- **Identification of professional gaps in own carrier**
  - ✓ Key terms definition
    - Professional gap
    - Labor market
  - ✓ Labor market Demand
    - Labor market role in training
    - Labor market opportunities
  - ✓ Techniques of identifying professional gaps
    - SWOT Analysis
    - Benchmarking competences
    - Identify important skills

- **Establishment of strategies to fill the professional gap**

  - ✓ Strategies to fill the professional gaps
    - Online Training
    - IAP
    - Volunteering
    - Attending seminars
  - ✓ Market opportunity
  - ✓ Matching strategies with labor market opportunities
    - Identifying occupational trends
    - Anticipating and matching skills and jobs
  - **Selection of relevant placement**
    - ✓ Identifying IAP stakeholders
      - Relevant workplaces
    - ✓ Responsibilities of the industrial attachment's stakeholders
      - Responsibilities of trainer
      - Responsibilities of learner
      - Responsibilities of host Industry
      - IAP placement process
      - Select Relevant workplaces

Resources required for the indicative content	
Equipment	• Computer, , projector, White/chalk board, Flipchart stand
Materials	• Papers, flip chart**,** pens / pencil, stickers, scotch tape
Tools	Scotch tape, Logbooks template form, Reference books, Evaluation form template
Facilitation techniques	• Individual research • Group discussion and presentation
Formative assessment methods	• Written assessment • Oral presentation

Learning outcome 4. Enhance acquired competencies at the workplace	**Learning hours: 170**
Indicative content	

- **Performing assigned tasks.**
  - ✓ Development of competencies related to one's field
    - Workplace safety and security activities

- Various planned task related to one's field
- Meetings with the internship supervisor

- **Preparation of IAP documents**

  ✓ Trainer and Trainee's documents
  - Logbooks
  - IAP report template
  - IAP Code of conduct

  ✓ Company's documents
  - Company supervisor logbook
  - Attachment report
  - General departmental information of companies
  - Training assessment forms

- **Description of work experience gained during IAP**
  ✓ Description of IAP work carried out
  ✓ Description of the competences acquired during IAP
  ✓ Description of challenges encountered in IAP
  - Technical challenges
  - Social Challenges

  ✓ Explanation of the approaches used to address challenges
  ✓ Clarification of recommendations toward
  - Company
  - College

- **Description of skills transfer**
  ✓ Definition of skills transfer
  ✓ Explanation of types of transferable skills
  - Social skills
  - Professional skills

- **Descriptions of utilization of skills acquired during IAP**
  ✓ Work experiences related to one's field

Resources required for the indicative content	
Equipment	• Equipment available at the workplace
Materials	• Raw/consumable materials at the workplace

Tools	• Vocational tools available at the workplace
Facilitation techniques	• Demonstration and simulation • Individual and group work • Trainer guided • Four step method • Developing method • Technical discussion
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

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